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The AMS-02 detector on the ISS - Status and highlights after 11 years on orbit

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Abstract. The Alpha Magnetic Spectrometer, AMS-02, is a magnetic spectrometer detector operating on the International Space Station (ISS) since May the 19**th**, 2011. More than 200 billion events have been collected by the instrument in the first 11 years of data taking, providing detailed and novel insights on the composition and energy spectra of cosmic rays up to TeV energies. We review the most recent AMS-02 measurements and the advances in the understanding of cosmic ray origin, acceleration and propagation physics.

1. The AMS mission

The Alpha Magnetic Spectrometer (AMS-02) is a precision particle physics detector operating on the International Space Station (ISS) to run a long-term, unprecedented precision mission of direct cosmic ray (CR) measurements in space (Figure 1).

The main physics objectives of the AMS mission include: (i) measurements and search of the antimatter components of CRs, for the investigation of fundamental physics and astrophysics frontiers, such as the search for primordial antimatter and for the origin of dark matter; (ii) measurements of all species of CRs from Hydrogen (Z=1) up to Iron (Z=26) and beyond for the understanding of the mechanisms of origin, acceleration and propagation of CRs; (iii) precision measurements of long-term and short-term time evolution of CR fluxes for understanding the mechanisms of propagation of charged particles in the Heliosphere.

The AMS-02 detector is based on a magnetic spectrometer made of a permanent magnet coupled to precision particle tracking and time-of-flight detectors to measure the rigidity, charge and velocity of CRs, and is completed with ancillary detectors to improve the accuracy and particle identification capabilities of the instrument (see Section 2). AMS-02 has been deployed in space using the Space Shuttle Endeavour and installed on ISS on May 19**th**, 2011. Since then, AMS-02 is continuously operating and collecting CRs. At the end of 2019, the AMS-02 tracker thermal cooling system has been objective of an upgrade [1] that has required a set of 4 Extra Vehicular Activities (EVAs). The interventions were successful, and the Upgraded Tracker Thermal Control System (UTTPS) has enabled AMS-02 to efficiently continue its operations for the lifetime of the ISS throughout at least 2030.

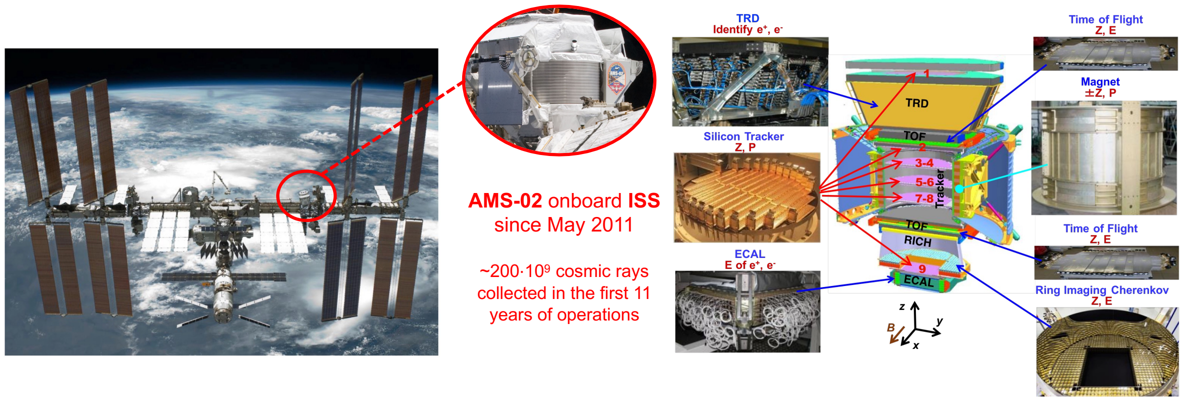

Figure 1. Left: AMS-02 on the ISS. Right: Representation of the AMS-02 instrument.

2. The AMS-02 instrument

The AMS-02 detector [2] consists of a spectrometer made of a 0.14 T dipolar permanent magnet coupled with a precision tracking system. Inside the magnetic volume and above and below the magnet, the 9 layers of the Silicon Tracker made of double-sided Si-μstrip sensors measure the particle crossing coordinates with a resolution of 5 μm for Z=6 and 10 μm for Z=1 particles over a lever arm of 3 m. Above and below the magnet, the Time of Flight (TOF) counters measure the particle velocity and trajectory direction. Anti-Coincidence Counters (ACC) surround the tracker in the magnet bore to identify and reject particles crossing outside the spectrometer field of view. The AMS-02 instrument is the most precise magnetic spectrometer operated in space, with capabilities of particle charge sign and rigidity measurements up to $\sim$2 TV for Z=1 and $\sim$3 TV for $Z > 1$ particles.

The AMS-02 particle identification capabilities are improved by additional subdetectors that provide redundant and complementary information to the magnetic spectrometer. The Transition Radiation Detector (TRD), located at the top of the instrument, identifies $e^{\pm}$ by the analysis of Transition Radiation X-rays in 20 layers of proportional tubes. The Ring Imaging Cherenkov detector (RICH), located below TOF, measures the particle velocity and charge by the analysis of the Cherenkov light produced by particles crossing its radiators. At the bottom of AMS-02, the 3D-imaging Electromagnetic Calorimeter (ECAL), made of Sci-fibers in a lead matrix, measures the topological development of showers in its 1296 cells over 17 radiation lengths (X_0) providing an energy measurement resolution better than 2% for energetic electrons and an electron/proton (e/p) separation capability $\sim 10^4$ that, combined with the independent e/p separation of TRD, allows to achieve a total e/p separation of $\sim 10^6$.

AMS-02 is equipped with a total of 300'000 electronics read-out channels which are processed by 650 microprocessors to provide the required data reduction for the transmission of data from AMS-02 on ISS to the ground with an average rate of 10 Mbit/s.

3. Scientific results of AMS-02

AMS-02 has collected $\sim 200{\cdot}10^9$ CRs since the start of its operations in 2011. The large amount of collected statistics and the accuracy in the data achieved by the complementary information provided by the 5 independent AMS-02 subdetectors has opened the opportunity for precision measurements of the chemical composition, energy spectra and time dependence of CR fluxes, usually reaching a detailed understanding of systematic uncertainties down to the level of O(%), which is typical of particle physics experiments at colliders and unprecedented in space particle measurements. Such accuracy has revealed new features in the fluxes of CRs, which were not

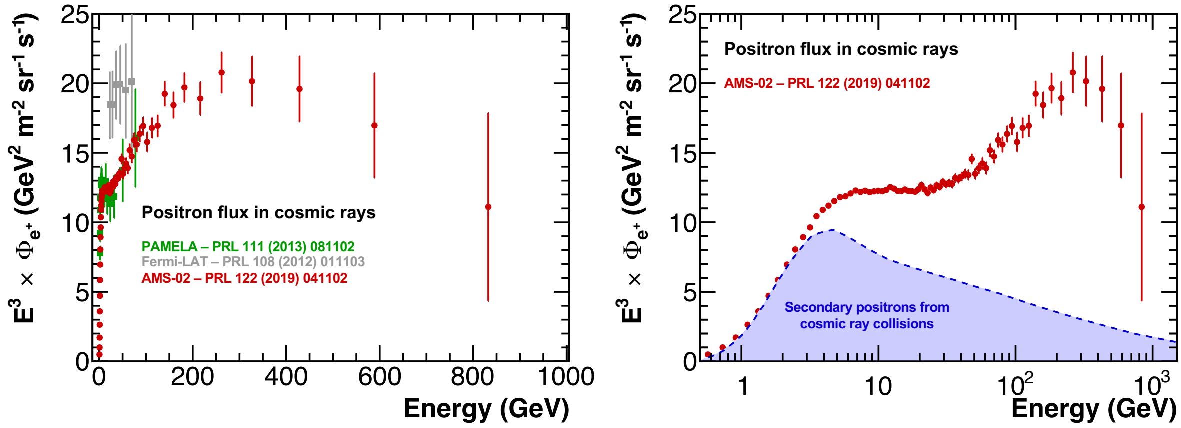

Figure 2. Left: Positron flux measurement in CRs by Fermi-LAT (gray) [5], PAMELA (green) [6], AMS-02 (red) [4]. The AMS-02 data significantly extend the measurements into the uncharted high energy region. Right: AMS-02 positron flux measurement compared to expected prediction from pre-AMS models of secondary positron production from CR collisions (blue) [7]. AMS-02 data show that an additional component of high energy positrons is required to describe the data. Figures readapted from [4].

expected in the models of origin, acceleration and propagation of CRs established before the AMS-02 era. The scientific community is now reinvestigating and revising the models in light of the recent AMS-02 results.

In the following we report a brief summary of the most representative results achieved by AMS-02. Please refer to [2] for a comprehensive review of AMS-02 measurements and results.

3.1. Measurements of the light antimatter components of cosmic rays

AMS-02 has measured the fluxes of electrons (e^-) and positrons (e^+) up to energies respectively of 1.4 TeV and 1 TeV [3, 4], extending by a factor ~ 4 the maximum energy reach measured by earlier satellite measurements (Figure 2). The AMS-02 measurements have shown, for the first time, the complex energy behaviour of the antimatter e^+ excess in CRs with evidence larger than $4\,\sigma$ of an energy cut-off at approximately 800 GeV. The AMS-02 measurement is compatible with a source term associated with secondary astrophysical e^+ produced in the collision of CRs, which dominates at low energies and predicted to be the only source of e^+ in standard models for CR origin, and a new additional source of e^+, which dominates at high energies. The data have risen widespread interest and discussions on the dominating source of primary high energy e^+, which could be explained in terms of annihilation of dark matter particles or acceleration of e^+ in astrophysical objects, and which is hardly consistent with secondary production of high energy e^+ in the interactions of CR nuclei with interstellar gas (see [3, 4] and references therein). The measured e^- flux features a different spectral shape than that of e^+. Contrary to the e^+ flux, the AMS-02 data show that at the $5\,\sigma$ level the e^- flux does not feature an energy cut-off up to 1.9 TeV. The different behaviour of the e^+ and e^- measured by AMS-02 is a clear evidence that most high energy e^- originate from different sources than high energy e^+.

The latest measurement of anti-protons ($\bar{p}$) from AMS-02 ranges from 1 to 500 GV [2], extending above the frontier of 150 GV reached by previous space experiments. The AMS-02 measurement of the anti-proton to proton ratio ($\bar{p}/p$), which flattens towards high energies, has shown evidence of a similar spectral behaviour between $\bar{p}$ and p up to 500 GV. This observation

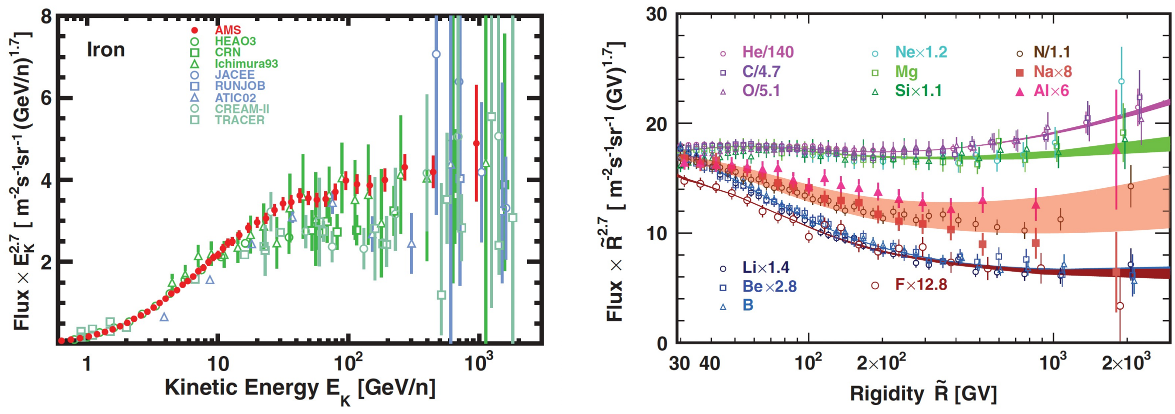

Figure 3. Left: Measurement of the Iron flux by AMS-02 (in red) compared with earlier space measurements (see [8] and references therein). Right: Cosmic nuclei fluxes measured by AMS-02 as a function of rigidity from $Z = 2$ to $Z = 14$ above 30 GV. Two classes of primary CRs, He-C-O and Ne-Mg-Si, and two classes of secondary CRs, Li-Be-B and F, are well distinct. N-Na-Al is a separate group, combination of primary and secondary CRs. For display purposes only, fluxes were rescaled as indicated. The shaded bands are to guide the eye. Figures from [8, 9].

Check what these classes are based on.

is difficult to explain in the scenario where, according to current models, anti-protons are purely secondary particles produced from collisions of CRs with the interstellar medium and for which the ($\bar{\mathrm{p}}$/p) ratio is expected to feature a decreasing trend with energy. The ($\bar{\mathrm{p}}$/p) ratio flattening is in fact at the limit of compatibility with the expected secondary $\bar{\mathrm{p}}$ production yield. In addition, some tantalizing hints of excesses revealed by the accuracy of AMS-02 data at low energies are also being discussed in terms of evidence of dark matter annihilation signatures (see [2] and references therein).

3.2. Measurements of cosmic ray nuclei

AMS-02 is producing precision results in the characterization of the properties of CR nuclei, from the lightest and most abundant protons to the heaviest Iron and trans-Iron elements. The large amount of statistics, the precise knowledge of the rigidity scale, the charge separation and the possibility to identify and correct for spurious fragmentation events in the detector materials have paved the way for a new era in the precision direct measurement of nuclei CRs in space.

Spectral features of CRs encode information on sources and propagation of CRs. AMS-02 measurements of CR nuclei are showing that the origin, acceleration and propagation mechanisms of CRs are more complex than what described by the established models, revealing unexpected features. Figure 3 shows a selection of results on CR nuclei flux measurements by AMS-02 and comparisons with earlier space experiments. The accuracy of the AMS-02 measurements has revealed, for the first time, the existence of at least two classes of primary CRs (He-C-O-Fe and Ne-Mg-Si) and two classes of secondary CRs (Li-Be-B and F). These classes have identical rigidity dependences in the same group, but differ in their features with other groups, and they all feature a hardening of their spectra above 200 GV. A similar hardening is also present in the proton flux, whose peculiar, different behaviour than all other CR nuclei has been characterized by the precise measurement of the p/He ratio. Finally, the analysis of AMS-02 data has provided evidence of a distinct third class of CR nuclei (N-Na-Al) which is well described by a combination of primary CR component and secondary CR component.

3.3. Long-term and short-term time evolution of cosmic ray fluxes

The large acceptance and high precision of AMS-02 open for accurate measurements of the low energy CR fluxes as function of time. This provides unique information to probe the dynamics of solar modulation, to allow the improvement of constraints for dark matter search, to investigate the processes of CR propagation in the Heliosphere, and to reduce the uncertainties in radiation dose predictions for deep space human exploration.

AMS-02 has provided the first and unique rigidity-resolved measurement of the fluxes of p and He first on a monthly base [10] and, very recently, on a daily base [11, 12] over a time period which nearly coincides to a full 11-year solar cycle, covering the ascending phase, the maximum, and descending phase to the minimum of solar cycle #24 (Figure 4). The data show peculiar long-term time dependences correlated to the 11-year solar activity cycle and short-term recurrent time variations with periods of 27, 13.5 and 9 days. Below 2.4 GV, an hysteresis between the He/p ratio and the He flux has been observed at the level of 7 σ, providing unexpected evidence that at low rigidities the modulation of He and protons are different before and after the solar maximum in 2014. In addition, time-resolved flux measurement for the e^- and e^+ fluxes [13] have shown that, besides coincident and expected common short-term features in both species, the long-term trend features a smooth transition after the polarity reversal of the solar magnetic field, showing a dependence that was not expected by the state-of-the-art models of CR propagation in Heliosphere [14, 15] and is providing for the first time evidence and accurate data to investigate the charge-dependent effects in solar modulation of CRs.

4. AMS mission: outlook and prospects

(2030)

AMS-02 is currently planned to be operated until the end of the ISS mission. To achieve a comprehensive understanding of CR origin, acceleration and propagation mechanisms, precision measurements of all CR species to describe unexpected observations based on most recent data are required and will be tackled by AMS-02. With improved collected statistics and understanding of detector systematic effects, the analysis of AMS-02 data will provide additional information with updates in maximum energy reach and accuracy on already published observables, such as light antimatter and nuclei, which could be useful to consolidate and possibly provide evidence of novel features in the CR fluxes.

No other magnetic spectrometer besides AMS-02 is currently planned to be operated in space, and its operations are expected to be, as of today, the only opportunity for a campaign of precision light antimatter CR measurements and heavy antimatter CR searches in space.

The continuous operations of AMS-02 will be also useful to monitor the flux of low-energy CRs over time completing the 11-year solar cycle and possibly extending the measurements through a complete 22-year solar magnetic field cycle with consistent data sets of precision data collected by the same instrument in the whole period. This is a unique opportunity to provide the largest data set of short-term, long-term, energy and charge-resolved CR flux measurements useful to investigate solar physics and the influence of the solar activity in the radiation environment of the Solar System, a field which is gaining increased momentum in view of radiation risk mitigation assessment for deep space human space exploration.

Check where in the cycle we are right now.

4.1. Acknowledgments

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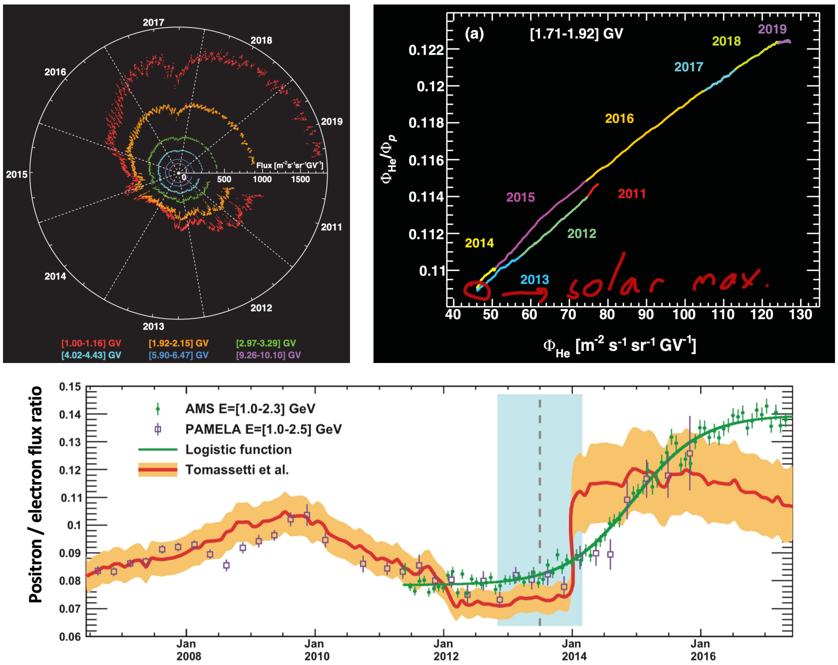

Figure 4. Daily AMS-02 proton fluxes measurements for six typical rigidity bins from 1.0 to 10.1 GV measured from May, 2011 to October, 2019 (Figure from [11]). Days with SEPs are removed for the two lowest rigidity bins. The proton fluxes, and similarly the He fluxes (see [12]), exhibit large variations with time, and the relative magnitude of these variations decreases with increasing rigidity. Top Right: Hysteresis before and after the solar maximum in 2014 between the He/p flux ratio as function of He flux from 2011 to 2019 in the [1.7-1.9 GV] rigidity bin (Figure from [12]). Bottom: e^+/e^- flux ratio measured before AMS-02 (PAMELA, violet [16]) and by AMS-02 (green) as function of time (Figure from [13]). The relative spectral behaviour shows a smooth transition after 2014, as consequence of changing charge-dependent modulation effects in the propagation in Heliosphere. The observed behaviour was not expected by the state-of-the-art models of CR propagation in Heliosphere (red line [14]).

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